

Possible problems with Hilbert transform

Arthur Gretton

22nd December 2005

We describe some reasons not to use the Hilbert transform, even for bandpass signals with narrow bands. We will see that the Hilbert transform gives the best answer *only* when the AM model is correct. In other circumstances, it gives a response that may be considered misleading. The latter is the case when recovering the envelope of bandpass filtered signals.

Figure 0.1: Imagine we have a sinusoidal signal at 300Hz modulating a carrier at 1000Hz. This can be expressed as follows: $y(t) = (\sin(2\pi f_e t) + d) \times \sin(2\pi f_c t)$, where f_e is the envelope frequency and f_c is the carrier frequency. The important thing to note here is the presence of the offset d , which *must* be large enough that the signal (in this case $\sin(2\pi f_e t)$) never crosses the zero axis. We now show an example where we set $d = 5$. Top left: true envelope plotted with the AM signal. Top right: spectrum of the AM signal. Bottom left: envelope and reconstructed estimate of the envelope. Bottom right: spectrum of the reconstructed signal.

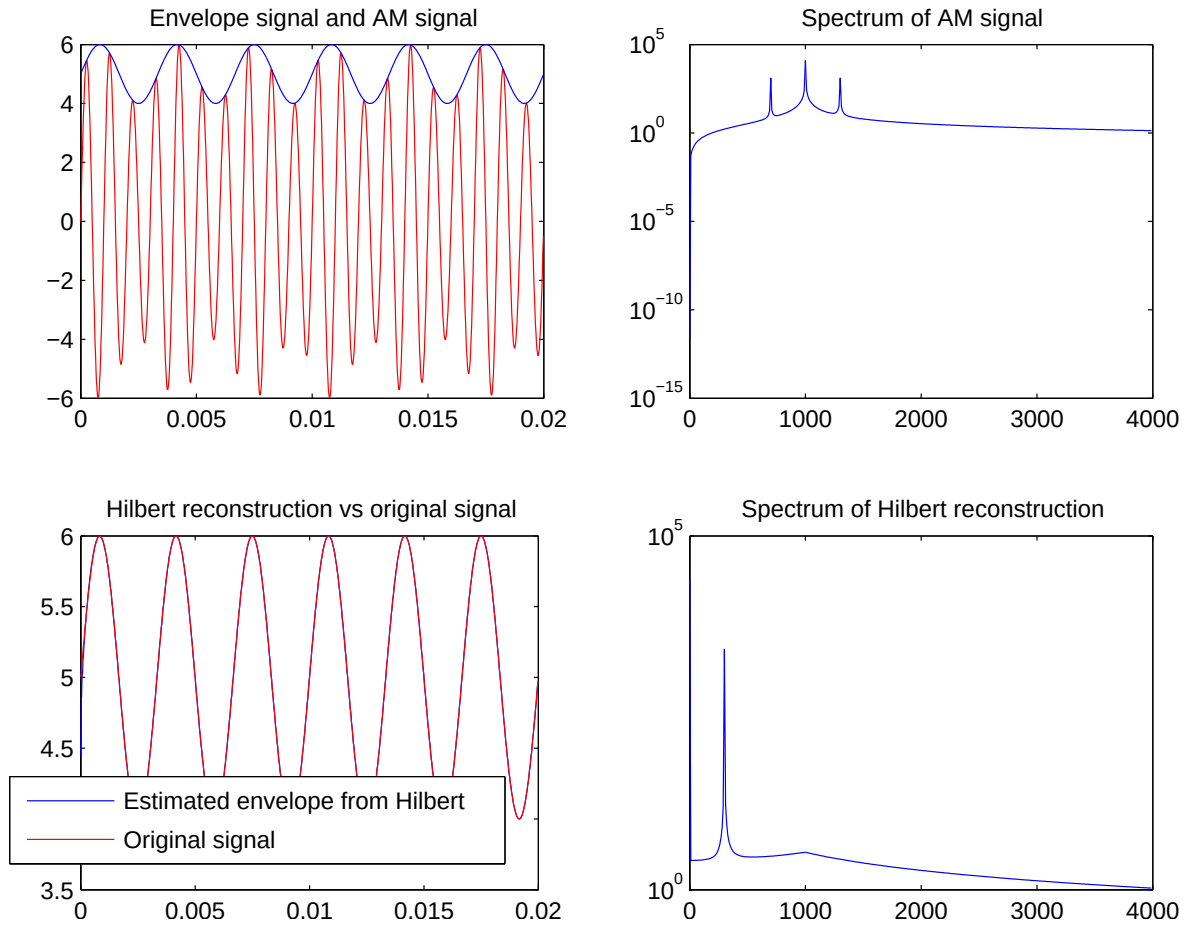


Figure 0.2: We now conduct the same experiment as above, but this time setting $d = 0.5$. We draw particular attention to the bottom left and bottom right plots. In the bottom left plot, we see the envelope "flips" at the point where the original signal crosses the zero axis. This creates a large number of high frequency components, which make no sense if our desire is to find a slow moving (low frequency) envelope modulating a high frequency carrier signal. Indeed, the "envelope" here contains strong frequency components that *exceed* the carrier frequency.

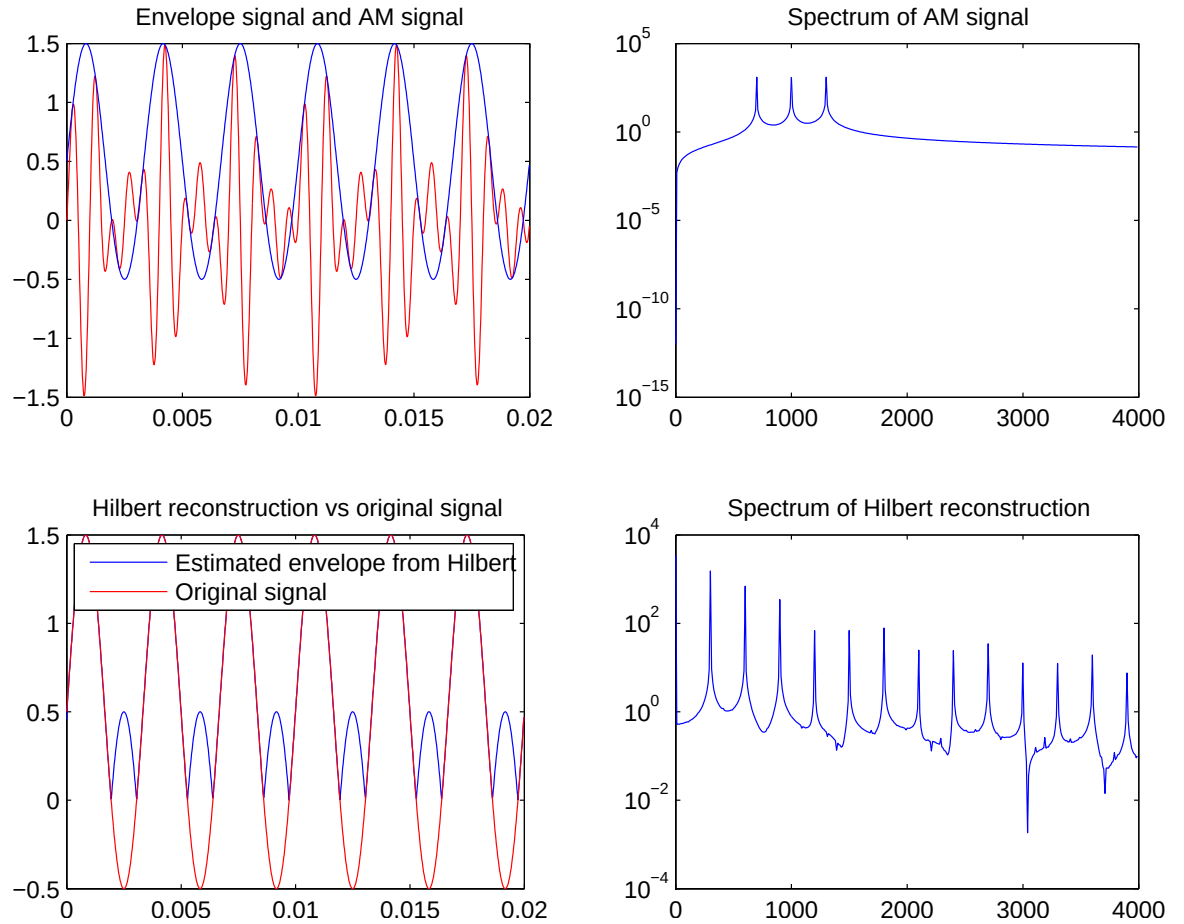


Figure 0.3: We now conduct the same experiment as above, with $d = 0.5$, using the absolute value + low pass filter approach. Here we see that we only recover the component of the envelope that varies slowly (by design of the low pass filter), which is truly an envelope (in that it contains no variation outside the frequencies that occurred in the passband of the original AM signal). There is some amplitude attenuation of the reconstructed envelope, which is likely due to the filter.

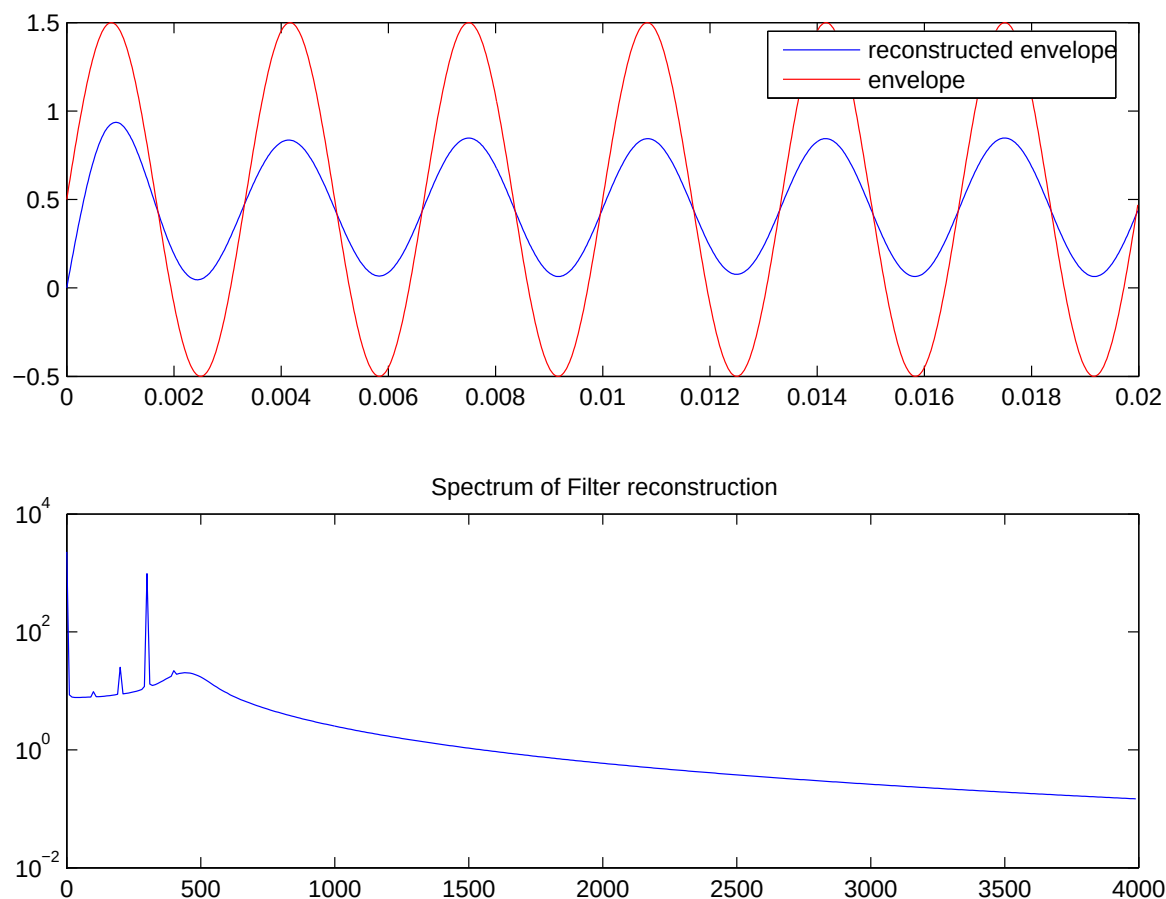


Figure 0.4: We next look at the case where we have a broad spectrum signal with dominant low frequencies, which is what we have from the electrodes. Hence this is a more realistic experiment from the point of view of the analysis we are conducting. The signal and its spectrum are plotted, as is the spectrum of a bandpass filtered version.

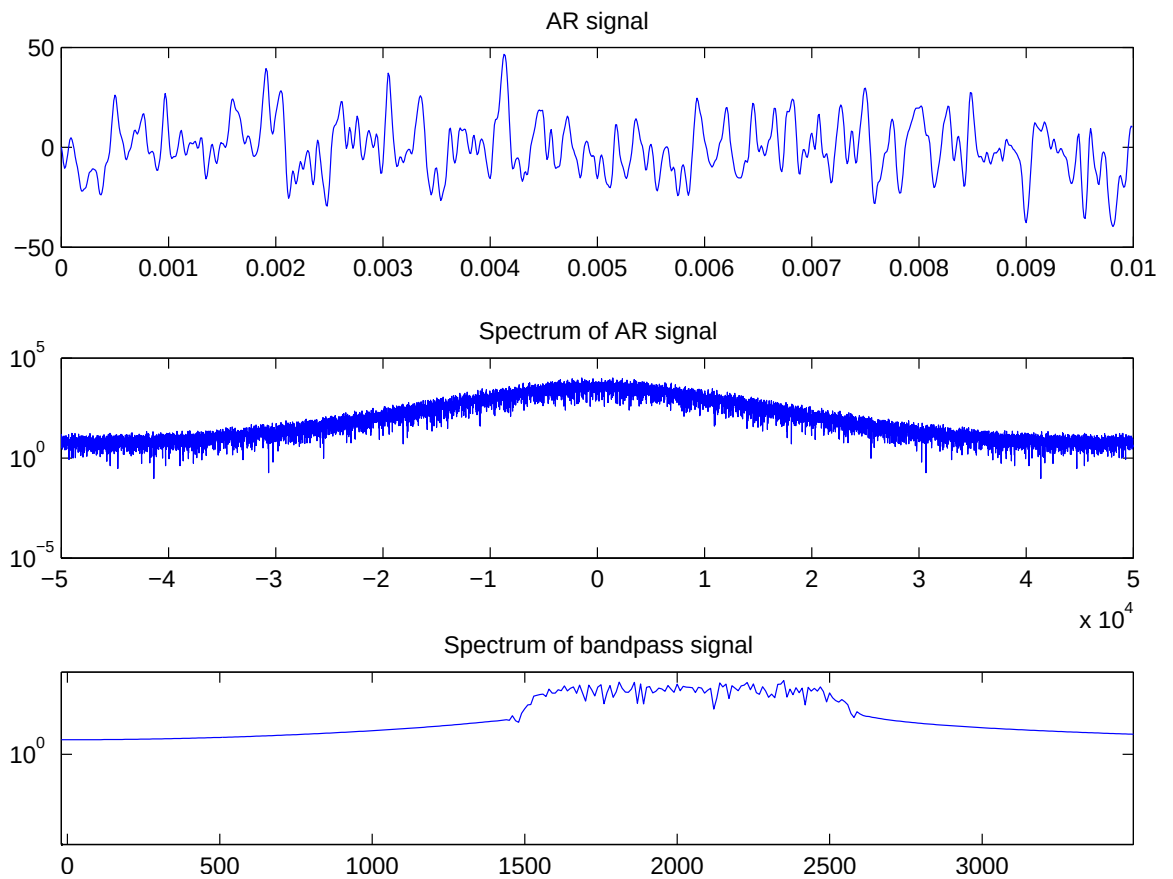


Figure 0.5: Finally, we look at the envelope recovered by the Hilbert transform and its spectrum, vs the envelope recovered by abs + low pass filter and its spectrum. The Hilbert case has an "envelope" that varies at close to the same time scale as the underlying carrier, and its spectrum is correspondingly broad (it has significant frequency components at higher frequencies than would be justified by the narrowness of the passband used in the bandpass filtering above). The abs+lowpass signal varies more slowly, and better reflects the envelope of the signal in that it has no components at frequencies higher than half the width of the passband (remember that by definition 0hz of the envelope is the middle of the passband).

